

Project Summary

Scope · business context · data model · data quality · report structure

Fictional company · synthetic seeded data · generated 2026-09-10

00 — Scope & Scaffold (Phase 0)

Project: SwiftBite Delivery — Marketplace Liquidity & the Zone-Hour Incentive Experiment **Fictional company:** SwiftBite Delivery — a Brazilian food-delivery marketplace, single metro. **Two-sided marketplace:** couriers = supply · orders = demand. **Date locked:** 2026-09-10

PT — resumo. Projeto de portfólio nº 4. Marketplace fictício de delivery de comida ("SwiftBite"), operando em **uma** cidade brasileira dividida em **12 zonas**, moeda **BRL apenas**. Em horários de pico e em zonas específicas, a oferta de entregadores não acompanha a demanda → ETA alto, cancelamento e entregadores ociosos na zona errada (liquidez ruim). Operações testou um **incentivo dinâmico por zona-horário** (bônus temporário pago ao entregador para entrar numa zona/janela) como **experimento randomizado por zona-dia**. A pergunta: o incentivo melhora a liquidez o suficiente para pagar o próprio custo, funciona só nas zonas carentes, e há canibalização das zonas vizinhas?

1. Premise

SwiftBite runs a food-delivery marketplace in one large Brazilian metro. It does not employ couriers; they log into the courier app and choose when and where to work. On a normal weekday afternoon the marketplace clears fine. But at **dinner peak on Friday and Saturday**, and in a handful of **structurally supply-short zones**, demand outruns the couriers available: order-to-courier assignment slows, **ETA climbs**, a wave of orders is **cancelled** (by the customer while waiting, or by SwiftBite when no courier can be assigned), and — the marketplace-liquidity signature — **couriers sit idle in other zones** at the same moment, logged in but without a run.

The Operations team piloted a **dynamic zone-hour incentive**: a temporary per-delivery bonus, announced in-app, paid to couriers who complete deliveries in a **named zone during a named peak block**. It was run as a **randomized experiment with the zone-day as the unit** (not the user, not the order): on a treated zone-day the incentive is armed for that zone's peak blocks; on a control zone-day it is not.

This project delivers a rigorous read-out of that experiment — **did the incentive buy enough liquidity to justify the bonus spend, is the effect concentrated in the supply-short zones, and does it cannibalise supply from neighbouring zones** — plus a Power BI report on marketplace health and courier economics.

2. Locked scope decisions

Dimension	Decision
Deliverable language	English (all docs, code, reports). Every deliverable also carries a short PT explanation of each step. Working chat in Portuguese.
Priority	Phase 10 (data analysis + the zone-hour incentive experiment) is the primary deliverable. Power BI track is secondary, scoped to 4 pages.
Geography / currency	One Brazilian metro (modelled on Belo Horizonte), 12 delivery zones, BRL only . No second city, no multi-currency — that ground is already covered by VoltEdge.
Data window	20 months: 2025-01-01 → 2026-08-31 . Baseline period = months 1–15; experiment sits in months 16–18; post-window observation to month 20.
Experiment window	10 weeks: 2026-04-06 → 2026-06-14 (window-months 16–18) — a shoulder stretch, no major holiday inside it. Post-window Jun 15 – Aug 31 reserved for persistence / courier-supply after-effects.
Unit of randomization	Zone-day (a zone × a calendar date). Secondary cut: zone × hour-block . Not user, order or courier.
ETL / data-quality depth	Balanced (matches the other three): ~6 simulated source systems, ~8 data-quality issue classes (\$5).
Marketplace scale	~ 12 zones ; ~ 3,500 couriers ever active, ~ 900 active in a typical week; ~ 14–16k orders/day metro-wide (~ 7.5M orders over the window).
Experiment scale	Peak-eligible zone-days only (dinner blocks, all days): 12 zones × ~70 window-days ≈ 840 zone-days , 50/50 treated/control, stratified by zone and by baseline supply-stress tier .
Power BI track	Semantic model as code (TMDL) + PBIR + <code>theme.json</code> + build script; 4 pages , one business question per page.
Visual identity	New palette, designed in Phase 5 — must not reuse AeroVanti (navy/amber), VoltEdge (navy) or Voxa+ (violet).

3. Phase-1 report pages (4 pages, one question each)

#	Page	The question it answers
1	Executive Overview	Is the marketplace healthy overall, and did the zone-hour incentive work well enough to keep?
2	Marketplace Health	<i>Where and when</i> does liquidity break — fulfillment rate, ETA, idle couriers vs unmet demand, by zone × hour?
3	Pricing & Incentive Experiment	Is the incentive effect real, significant and safe on the guardrails — including neighbour-zone cannibalisation — or is one zone carrying it?
4	Courier Economics	What do couriers earn, how utilised are they, and how elastic is courier supply to the incentive (R\$ of bonus per incremental delivered order)?

4. Experiment design (locked at Phase 0, detailed in Phase 1)

Element	Decision
Unit of randomization	Zone-day . On a treated zone-day, the dynamic incentive is armed for that zone's peak blocks (18:00–22:00). Secondary analysis at zone × hour-block .
Eligible units	Peak-eligible zone-days : every zone, every date in the 10-week window, restricted to the dinner peak (the only time the incentive is offered). ~840 zone-days.
Assignment	Stratified randomization : within each zone and each baseline supply-stress tier (short / balanced / long, fixed from months 1–15), zone-days are split 50/50 treated/control. No day is switched mid-day; no re-randomization.
Treatment	A per-delivery bonus (a flat R\$ amount, set in <code>etl/config.py</code>) for deliveries completed in the treated zone during the peak block , plus an in-app "surge zone" push to nearby couriers.
Control	No bonus, no push — normal marketplace.
Primary metric	Fulfillment rate = orders delivered ÷ orders placed, at the zone-day (peak-block) grain.
Guardrail metrics	(1) Incentive cost per incremental delivered order (BRL); (2) ETA p50 and p90; (3) cancellation rate (customer-, courier- and no-courier-initiated); (4) courier earnings per active hour ; (5) neighbour-zone spillover — fulfillment rate and idle-courier ratio in <i>adjacent, control</i> zones on treated zone-days (the cannibalisation check).
Analyses required	Balance / sample-ratio check at zone-day grain; stated hypothesis + MDE / power computed for the zone-day unit (small N, intra-zone correlation → cluster-robust SEs and a randomization-inference check); primary effect with a 95% CI; heterogeneity by baseline supply-stress tier and by zone , with a formal arm × tier interaction test ; spatial-spillover / cannibalisation estimate with a stated sign and bound; persistence in the post-window weeks.

Underlying true effect: decided during Phase 3 (ETL) and baked into the generated data with plausible noise, a modest **neighbour-zone cannibalisation** term, and heterogeneity by supply-stress tier. The true values live only in `etl/config.py`; Phase 10 must detect and quantify them as if blind.

5. Data-quality issues to inject (balanced set)

1. **Multiple date/time formats** across source exports — DD/MM/YYYY, ISO, and Unix-epoch milliseconds in the GPS ping log.
2. **Timezone drift** — one source (payments) logs in UTC while everything else is `America/Sao_Paulo`; no offset column.
3. **Money as strings** — `"R$ 12,90"`, `"12.90"`, `"1.234,56"` mixed in payout and basket-value fields.
4. **GPS noise & out-of-area pings** — lat/long jitter, a tail of pings hundreds of km away (device cold-start), and (0, 0) nulls.
5. **Duplicate rows** — the consumer app retries on flaky mobile connections, so a slice of `orders` and `order_status_events` rows are exact or near-exact duplicates.
6. **Missing master rows** — some `courier_id`s in the trip log have no row in the courier dimension export (referential gap).
7. **Zone boundary redraw** — SwiftBite re-cut two zone polygons at window-month 9; a **current** zone-assignment table disagrees with the **as-booked** zone on older orders.
8. **Mojibake / mixed encoding** in customer names, address strings and restaurant names.

6. Simulated source systems

System	Feeds	Grain
Consumer order app	<code>orders</code> , <code>order_status_events</code> , basket value, address → zone	order / status event
Courier app	<code>courier_sessions</code> (login/logout), <code>shift_zone_blocks</code> , offer → accept/reject/timeout	session / offer
GPS / location service	<code>location_pings</code> (courier position over time), used to derive idle vs en-route minutes	ping (~30–60 s)
Dispatch / assignment engine	offer→assignment→pickup→dropoff timestamps, assignment latency, "no courier found" events	assignment
Payments & payouts	order payment, courier payout, incentive bonus payout	transaction
Incentive / campaign system	<code>incentive_campaigns</code> , per zone-day arm + bonus params + spend	zone-day
Zone / geo reference	zone polygons, adjacency graph, area, baseline supply-stress tier	zone (+ history)

7. Out of scope (explicit)

- A second city, region expansion, multi-currency / FX (covered by VoltEdge).
- Restaurant-side economics (menu pricing, commission take-rate optimisation), except as an input to demand.
- Consumer-side pricing / promo experiments (delivery fee, discount coupons) — the intervention studied here is **supply-side**.
- Real-time dispatch-algorithm design; routing optimisation.
- A courier-churn *prediction* model — Phase 10 quantifies supply elasticity and the experiment effect; it does not ship a scorecard.
- Fraud (GPS spoofing, incentive gaming) beyond a noted data-quality tail.
- Real personal data — every courier, customer, order, ping and payout is synthetic and seed-generated.
- The **true incentive effect size and cannibalisation term** — decided in Phase 3, not here.

8. Resolved-assumptions log

#	Assumption	Resolution
A1	City & geography	One metro (modelled on Belo Horizonte), 12 zones , adjacency graph fixed in the geo reference (with one mid-window redraw as a DQ issue).
A2	Window & anchor	20 months, 2025-01-01 → 2026-08-31; data-gen date 2026-09-10.
A3	Baseline supply-stress tier	Each zone tagged short / balanced / long on supply, fixed from months 1–15 behaviour; used for stratification and heterogeneity.
A4	"Active courier"	≥ 1 completed delivery in the trailing 7 days (weekly) / trailing 28 days (monthly cut).
A5	Liquidity metric	Available courier-hours ÷ demanded delivery-hours, by zone × hour; < 1 = supply-short.
A6	Fulfillment rate	Orders delivered ÷ orders placed, same zone-hour; the experiment's primary metric.
A7	ETA	Order-placed → delivered wall-clock minutes; report p50 and p90 (p90 is the customer-pain metric).
A8	Cancellation taxonomy	customer-initiated · courier-initiated (after accept) · no-courier-found (system).
A9	Experiment unit / split	Zone-day , 50/50, stratified by zone × baseline supply-stress tier; peak block 18:00–22:00.
A10	Experiment window	10 weeks, 2026-04-06 → 2026-06-14 (window-months 16–18); post-window to 2026-08-31.
A11	Primary metric & decision rule	Fulfillment rate; ship if lift is positive, significant at $\alpha = 0.05$ (cluster-robust), ≥ MDE, and incentive cost per incremental delivered order is below the stated threshold.
A12	Guardrails	Incentive cost / incremental order; ETA p50 & p90; cancellation rate; courier earnings / active hour; neighbour-zone spillover .

#	Assumption	Resolution
A13	MDE / power	Computed for the zone-day unit ($N \approx 420/\text{arm}$), with intra-zone ICC; randomization-inference cross-check. Planning values finalized in Phase 1.
A14	Cannibalisation	A real, modest negative spillover to adjacent control zones is baked into the data in Phase 3; Phase 10 must estimate its sign and size.
A15	Currency	BRL only; no FX.
A16	True effect	Not decided here; set in ETL (Phase 3), detected blind in Phase 10.

9. Folder scaffold (created)

```
SwiftBite-Delivery-Analytics/  
docs/          00 scope · 01 business context & KPIs · 02 architecture · 03 data dict ·  
               04 ETL · 05 model · 06 DAX · 07 visual identity · 08 data quality ·  
               09 report structure · walkthrough.md · video script · docs bundle  
etl/           config (one SEED) · gen/ · 01 generate raw · 02 clean/stage ·  
               03 build curated · dq_checks · doc + DAX generators · run_pipeline  
data/  
  raw/         source-faithful messy exports          (git-ignored, regenerated)  
  staging/     typed stg_* tables                      (git-ignored, regenerated)  
  curated/     Kimball star, CSV + Parquet              (git-ignored, regenerated)  
  quality/     dq_report.md  
powerbi/       semantic model as code (TMDL) · PBIR report · theme · build script  
assets/        brand.py · logo / page-background generators  
data_analysis/ sql/ python/ findings/ parity/ deliverables/ outputs/ ← PRIMARY  
portfolio/     5 gallery images only  
README.md
```

End of Phase 0 deliverable. Next gate: Phase 1 (docs/01_business_context_kpis.md) — drafted alongside this; review both before Phase 2.

SwiftBite Delivery — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-10 · **Marketplace-history window:** 2025-01-01 → 2026-08-31 (20 full months). **Embedded experiment:** the zone-hour incentive test, 2026-04-06 → 2026-06-14 (10 weeks, window-months 16–18). **Audience:** SwiftBite's Operations, Growth and Finance leads.

This document frames the problem and defines the metrics. It deliberately does **not** state whether the zone-hour incentive worked, or by how much — that is the job of the read-out (Phase 10). Everything here was knowable *before* the analysis. The true underlying treatment effect — including the neighbour-zone cannibalisation term — is decided during ETL (Phase 3), baked into the synthetic data with noise and heterogeneity, and kept only in `etl/config.py`; Phase 10 must detect and quantify it as if blind.

PT — o que este documento é. Enquadra o problema de negócio e define as métricas de um marketplace de delivery. **Não** diz se o incentivo por zona-horário funcionou — isso é a Fase 10. O efeito verdadeiro (e a canibalização entre zonas vizinhas) é plantado nos dados sintéticos na Fase 3 e fica só em `etl/config.py`; a análise tem que recuperá-lo "às cegas".

1. The situation

PT. SwiftBite é um marketplace de delivery de comida numa única cidade grande do Brasil. Não emprega entregadores — eles entram no app e escolhem quando e onde rodar. Em **pico de jantar de sexta e sábado**, e em algumas **zonas cronicamente com pouca oferta**, a demanda passa da oferta: a fila de atribuição atrasa, o **ETA sobe**, uma onda de pedidos é **cancelada**, e — a assinatura de liquidez ruim — há **entregadores ociosos noutras zonas** ao mesmo tempo. Operações testou um **bônus dinâmico por zona-horário** como experimento randomizado por **zona-dia**.

SwiftBite Delivery operates a food-delivery marketplace in one large Brazilian metro (modelled on Belo Horizonte). It is a **two-sided marketplace**:

- **Demand** — consumers place orders in the SwiftBite app; each order belongs to a **delivery zone** (from the delivery address) and a **time block**.
- **Supply** — independent couriers log into the courier app and choose their hours and rough working area. SwiftBite does not roster them; it can only *influence* where supply goes — with information, with assignment logic, and with money.

Most of the week, the metro clears fine: a courier is assigned in a minute or two, ETA is 30–40 minutes, almost every order is delivered. The problem is **concentrated in time and space**:

1. **Peak dinner on Friday and Saturday** (≈ 18:00–22:00) — order volume roughly triples over a weekday lunch, but courier supply rises far less.
2. **A handful of structurally supply-short zones** — outer-ring residential zones with long trips, poor return-load prospects and few couriers who live nearby.

When those overlap, three things happen together and define **poor marketplace liquidity**:

Symptom	Metric it shows up in
Assignment slows / fails	Assignment latency ↑, "no courier found" cancellations ↑
Customers wait, then bail	ETA p90 ↑, customer-initiated cancellation rate ↑
Supply is in the wrong place	Idle-courier ratio ↑ in <i>other</i> zones at the same minute
Orders never delivered	Fulfillment rate ↓ (the primary experiment metric)

The Operations team's hypothesis: a **temporary, targeted per-delivery bonus** — announced in-app for a **named zone during the peak block** — will pull enough nearby couriers into that zone to lift fulfillment, at a bonus cost lower than the contribution margin of the orders it saves. Growth's worry: the couriers who move are **taken from adjacent zones**, so the local win is partly a transfer, not new supply. Finance's worry: the bonus is paid on **every** delivered order in the zone-block, including the ones that would have happened anyway. The experiment is designed to answer all three.

2. Company profile

Name	SwiftBite Delivery (legal entity <i>SwiftBite Tecnologia e Logística Ltda.</i>)
Headquarters	Belo Horizonte (MG), Brazil
Business	On-demand food-delivery marketplace — consumers ↔ restaurants ↔ independent couriers
Footprint in scope	One metro, 12 delivery zones. ~6,000 active restaurants.
Age	Founded 2022; this project's 20-month window is 2025-01 → 2026-08 (a mature, steady operation, not a launch ramp).
Currency	BRL only. Single-country, single-city; no FX in scope.
Reporting calendar	Calendar months for finance; operations reporting is by day and by hour-block ; the experiment is analysed by zone-day.

Scale at window end	~ 3,500 couriers active over the window (~ 900 in a typical week); ~ 15,000 orders/day metro-wide (~ 7.5M orders across the window); average basket ~ R\$ 62 ; average delivery fee ~ R\$ 8.50 ; SwiftBite take rate ~ 22% of basket + fee share.
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Marketplace positioning

SwiftBite is the **#2 player** in this metro behind a national incumbent. It competes on **ETA reliability** and **courier earnings** (couriers multi-app; the platform with better peak earnings gets their hours). Restaurant commission is not a lever in this project — it is treated as a fixed input to unit economics.

3. The two sides of the marketplace

PT. Os dois lados nunca devem ser misturados na análise: **oferta** = horas de entregador disponíveis; **demand** = pedidos (ou horas-entrega demandadas). Liquidez é a razão entre os dois num dado par zona-hora.

3.1 Supply — couriers

Concept	Definition
Courier session	A continuous logged-in period in the courier app (login → logout).
Active minutes	Session minutes assigned to an offer or en route (offer→accept→pickup→dropoff).
Idle minutes	Session minutes logged in but not on a run and not in cooldown. High idle = supply present but unproductive (often in the wrong zone).
Available courier-hours (a zone-hour)	Σ session-hours whose position is in the zone during the hour, net of cooldown. The supply quantity.
Active courier	≥ 1 completed delivery in the trailing 7 days (weekly cut) / 28 days (monthly cut).
Courier home zone	Modal drop-off zone over the trailing 28 days — used only for descriptive supply geography, never for assignment.
Multi-apping	Assumed but not observed directly; shows up as couriers going idle-then-logout during a SwiftBite lull. A stated limitation.

3.2 Demand — orders

Concept	Definition
Order	A placed consumer order with a delivery address → delivery zone and a time block .
Demanded delivery-hours (a zone-hour)	Orders in the zone-hour × the period's average service time (pickup + drive + drop). The demand quantity, in the same unit as supply.
Order outcome	delivered · cancelled_customer · cancelled_courier · cancelled_no_courier · cancelled_restaurant .
ETA	Order-placed → delivered wall-clock minutes. Reported p50 (typical) and p90 (the customer-pain tail).
Assignment latency	Order-ready → courier-assigned seconds.

3.3 Liquidity — where the two meet

Metric	Definition	Read
Liquidity ratio	available courier-hours ÷ demanded delivery-hours, per zone × hour	< 1 → supply-short; the marketplace cannot clear without ETA blowing out or orders dropping
Fulfillment rate	orders delivered ÷ orders placed, same zone-hour	the headline outcome — and the experiment's primary metric
Idle-courier ratio	idle courier-hours ÷ available courier-hours, per zone × hour	high and low fulfillment in a <i>neighbouring</i> zone at the same time = a spatial mismatch the incentive is meant to fix
Unmet demand	placed – delivered, per zone-hour	the volume the incentive is trying to recover

4. Zones & geography

PT. 12 zonas cobrindo a cidade; cada zona tem vizinhas (grafo de adjacência) e um **tier de estresse de oferta** (curta / equilibrada / longa) fixado pelo comportamento dos meses 1–15. O tier é usado para estratificar o experimento e para a análise de heterogeneidade. A canibalização é justamente supor que o incentivo numa zona rouba oferta das **zonas adjacentes**.

Property	Detail
Zone count	12, tiling the metro; each is a group of neighbourhoods with one dispatch centroid.
Adjacency graph	Fixed in the geo reference; each zone has 2–5 neighbours. Cannibalisation is defined against adjacent zones .

Property	Detail
Baseline supply-stress tier	Each zone tagged short / balanced / long from months 1–15 liquidity (4 short · 5 balanced · 3 long, indicative). Fixed pre-experiment; used for stratification and heterogeneity.
Zone boundary redraw	Two zone polygons were re-cut at window-month 9 (Sep 2025) — a deliberate data-quality issue: the current zone map disagrees with the as-booked zone on older orders (§ data-quality doc).
Trip geography	"short" zones have longer average trips and lower return-load probability — the structural reason supply avoids them.

5. The zone-hour incentive experiment

5.1 Design

PT. Unidade de randomização = **zona-dia** (uma zona num dia). Em zona-dia tratado, o bônus por entrega fica ligado no bloco de pico (18:00–22:00) daquela zona; em controle, não. Split 50/50, **estratificado por zona e por tier de estresse de oferta**. Nada muda no meio do dia; sem re-randomização.

Element	Decision
Unit of randomization	Zone-day — one zone on one calendar date. Secondary analysis at zone × hour-block .
Eligible units	Peak-eligible zone-days: every zone, every date in the 10-week window, peak block 18:00–22:00 (the only time the incentive is ever offered). ≈ 840 zone-days (12 × 70).
Assignment	Stratified randomization : within each zone and each baseline supply-stress tier , zone-days split 50/50 treated/control, balanced across day-of-week (so treated and control both get their share of Fri/Sat). Assignment fixed at 00:00 local; no mid-day switch; no re-randomization.
Treatment	A flat per-delivery bonus (R\$ amount set in <code>etl/config.py</code>) for orders completed in the treated zone during the peak block , plus an in-app "surge zone" push to couriers within ~2 km.
Control	No bonus, no push — the normal marketplace.
Blinding	Couriers see the bonus (it must be visible to work); they do not see the experiment. Customers see nothing.
Duration	10 weeks, 2026-04-06 → 2026-06-14. Post-window observation to 2026-08-31 for persistence.

5.2 Hypotheses

- **Business hypothesis.** Arming a targeted peak-block bonus in a zone raises **fulfillment rate** in that zone-block (more couriers present → fewer no-courier cancellations, lower ETA, fewer customer bail-outs), at a **bonus cost per incremental delivered order below the contribution margin of a saved order**, and the effect is **largest in supply-short zones**. A secondary hypothesis is that some of the gained supply is **drawn from adjacent zones** (cannibalisation), so the metro-level net effect is smaller than the sum of local effects.
- **Statistical hypothesis (primary).** H_0 : fulfillment rate is equal on treated and control zone-days over the window. H_1 : treated > control. Tested one-sided at $\alpha = 0.05$ with **cluster-robust (zone) standard errors**; a two-sided 95% CI on the difference is always reported, alongside a **randomization-inference** p-value as a small-N cross-check.

5.3 Metrics

Role	Metric	Definition	Decision rule
Primary	Fulfillment rate	Orders delivered ÷ orders placed, treated zone-days vs control zone-days, peak block.	Keep the incentive if lift is positive, significant ($\alpha = 0.05$, cluster-robust), $\geq$ the MDE, and guardrail G1 passes.
Guardrail G1	Incentive cost per incremental delivered order (BRL)	Total bonus spend ÷ (incremental delivered orders vs control).	Ship only if below the contribution margin of a delivered order (planning value ≈ R\$ 9–11). Reported with a CI.
Guardrail G2	ETA p50 / p90 (min)	Placed→delivered minutes, by arm.	Directional; flag if treated is worse (should improve or hold).
Guardrail G3	Cancellation rate	Cancelled ÷ placed, split by cause (customer / courier / no-courier).	Flag if total or no-courier rate is worse on treated.
Guardrail G4	Courier earnings per active hour (BRL)	Payout incl. bonus ÷ active hours, couriers working the zone-block.	Context + fairness; expected up on treated (that is the mechanism).
Guardrail G5	Neighbour-zone spillover (cannibalisation)	Δ fulfillment rate and Δ idle-courier ratio in adjacent control zones on treated zone-days vs non-treated-neighbour zone-days.	Estimate sign and size ; flag if the metro-level net lift is < X% of the summed local lift.

5.4 Power & minimum detectable effect (zone-day unit)

PT. A unidade é **zona-dia**, então o N efetivo é pequeno (~420 por braço) e há **correlação intra-zona** — o cálculo de poder tem que usar essa unidade, não o número de pedidos. Erros-padrão com cluster por zona + inferência por randomização.

Assumption	Value (planning)
Analysis unit	zone-day (peak block)
Zone-days per arm	≈ 420 (≈ 840 eligible, 50/50)
Control fulfillment rate at peak, supply-short zones (planning value)	≈ 0.86
Between-zone-day SD of fulfillment rate (peak)	≈ 0.07 (finalized from realized baseline in Phase 10)
Intra-zone correlation (ICC)	assumed ≈ 0.15 → design effect ≈ 1.9 for ~35 days/zone/arm
α (two-sided) / power	0.05 / 0.80
MDE on fulfillment rate , all zones pooled	≈ +1.3 pp
MDE within the supply-short tier (≈ 140 zone-days/arm)	≈ +2.2 pp — the tier the decision hinges on; adequately but not generously powered
MDE within a single zone	≈ +4–5 pp — single-zone effects are directional only
Secondary unit (zone × hour-block)	~4× the rows, used for the novelty/decay and time-of-peak cut, not for the headline test

The power calculation is re-run from realized pre-period variance and reproduced in code in Phase 10; the values above are the planning basis.

5.5 Threats to validity the read-out must address

Threat	How it is handled
Assignment imbalance / SRM	χ^2 on treated/control counts overall, per zone, per supply-stress tier and per day-of-week; investigate if $p < 0.01$.
Covariate imbalance	Standardized differences on pre-period fulfillment, ETA, order volume, liquidity ratio, weather flag, day-of-week; Love plot.
Spatial spillover / cannibalisation (SUTVA violation)	This is a <i>studied effect</i> , not just a nuisance: G5 estimates it directly by comparing adjacent-control zones on treated vs non-treated-neighbour days. The primary estimate is also reported excluding zone-days adjacent to a treated zone, as a bound.
Novelty / courier learning	Weekly treated-vs-control lift across the 10 weeks; test for a decaying trend (weeks 1–3 vs 8–10); report the settled lift.
Weather & events	Rain and city-event flags as covariates; control zone-days are concurrent so metro-wide shocks difference out, but zone-local rain does not — hence the covariate.
Demand endogeneity	The bonus could itself change customer behaviour (faster ETA → more orders). Fulfillment rate is a ratio, but order <i>volume</i> is also reported by arm so a demand response is visible, not hidden.
Small number of clusters	12 zones → cluster-robust SEs are anti-conservative; a randomization-inference p-value and a wild-cluster bootstrap are reported alongside.
Multiple comparisons	One primary metric fixed in advance; 5 guardrails + the tier/zone heterogeneity cut are secondary and CI-reported with a family-wise-error note.
Persistence	The decision is on the in-window settled lift; the post-window weeks are reported as a persistence signal, not a second test.

5.6 Heterogeneity analysis (required)

Effect on fulfillment rate estimated **within each baseline supply-stress tier** (short / balanced / long) and **within each zone**, with per-cell CIs and a formal **arm × tier interaction** test. The prior to test — not assume — is that the effect is **largest in supply-short zones** (most unmet demand to recover) and **near zero in supply-long zones** (already clearing; the bonus is pure cost there). The operational decision — roll the incentive out everywhere, or only to short zones — hinges on this cut and on G1 computed per tier.

6. The analytics window — why 20 months

Anchor	Value
Window length	20 complete months: month 1 = Jan 2025 , month 20 = Aug 2026
Baseline period	months 1–15 (Jan 2025 – Mar 2026) — defines each zone's supply-stress tier
Experiment	window-months 16–18 → 2026-04-06 → 2026-06-14 (10 weeks)
Post-window observation	2026-06-15 → 2026-08-31 (~11 weeks) — persistence
Year-over-year	months 1–12 vs a trailing-12 ending at month 20, for the report's trend context only

Why 20 months. Long enough for two full seasonal cycles of the parts that matter (Carnaval, June festas, school holidays, rainy season) and a stable 15-month baseline before the experiment; the operation is mature over the whole window, so there is **no launch-ramp comparability caveat** (unlike AeroVanti). Shorter than VoltEdge's 24 because there is no multi-year financial trend to establish — the value is in the experiment.

7. Seasonality & patterns embedded in the data

Pattern	Where it shows
Weekly — Fri/Sat dinner peak	Order volume ~3× a weekday lunch; the peak block the experiment targets
Daily — lunch (11:30–14:00) and dinner (18:00–22:00) double peak	Dinner is longer, more supply-short
Rainy season (Oct–Mar in the Southeast)	Demand ↑ (people don't go out), courier supply ↓ (fewer want to ride) → the worst liquidity of the year; the experiment's first 2 weeks catch the tail of it
Carnaval (Feb/Mar)	Multi-day demand collapse then rebound — outside the experiment window by design
June festas / school July holidays	Elevated weekend demand; overlaps the experiment's back half — handled by the concurrent control arm
Payday (5th & 20th)	Mild order-volume and basket-size lift
Big football matches / shows	Zone-local demand spikes; flagged as an event covariate
Courier supply drift	Slow metro-wide decline in couriers/week across 2026 as a competitor raises pay — a backdrop trend, differenced out by the concurrent control

8. Stakeholder personas → report pages

PT. Três personas, uma pergunta permanente cada — mais a pergunta executiva. As quatro viram as quatro páginas do relatório.

#	Persona	Standing question	Page
1	Head of Operations	Is the marketplace healthy this week, and did the zone-hour incentive earn its cost?	Executive Overview
2	Marketplace / Liquidity Manager	Exactly which zones and hours are supply-short, how bad is the ETA and cancellation damage, and where is idle supply sitting while that happens?	Marketplace Health
3	Growth / Courier Supply Lead (<i>and the experimentation partner</i>)	Is the incentive effect real, significant and safe on the guardrails — including cannibalisation of neighbour zones — or is one zone carrying it?	Pricing & Incentive Experiment
4	Finance / Unit Economics	What do couriers earn per hour, how utilised is supply, and how many reais of bonus buy one extra delivered order?	Courier Economics

9. KPI framework

Benchmark ranges are **directional**, drawn from public delivery-marketplace disclosures and industry commentary (gig-economy and last-mile studies), adapted to a single-metro #2 player. They set targets; they are not claims about named competitors.

9.1 Marketplace health & demand

KPI	Definition	Benchmark	Target
Orders placed / day	Distinct placed orders	—	on plan
Fulfillment rate %	Delivered ÷ placed	95%–99%	≥ 98% metro; ≥ 95% in every zone-hour
ETA p50 / p90 (min)	Placed → delivered	p50 28–38 · p90 45–70	p90 ≤ 55
Cancellation rate %	Cancelled ÷ placed	2%–6%	≤ 3%
— of which no-courier %	<code>cancelled_no_courier</code> ÷ placed	0.3%–2%	≤ 0.5%
Assignment latency p90 (s)	Ready → assigned	60–240	≤ 150
Repeat-order rate % (28d)	Customers with ≥ 2 orders in 28d	—	context

9.2 Supply — couriers

KPI	Definition	Benchmark	Target
Active couriers / week	≥ 1 delivery in trailing 7d	—	≥ plan
New couriers / month	First delivery in the month	—	≥ churn
Courier CAC (BRL)	Acquisition + onboarding-bonus spend ÷ new activated couriers	—	≤ plan
Courier 4-week retention %	Active in week 4 after first delivery	45%–70%	≥ 60%
Available courier-hours / zone-hour	Σ session-hours in the zone-hour (net cooldown)	—	≥ demanded hours
Idle-courier ratio %	Idle ÷ available courier-hours	15%–35%	≤ 25% (and not concentrated)
Deliveries per active hour	Completed deliveries ÷ active hours	1.6–2.6	≥ 2.0 at peak

9.3 Liquidity (zone × hour — the core of page 2)

KPI	Definition	Benchmark	Target
Liquidity ratio	available courier-hours ÷ demanded delivery-hours	≥ 1.0	≥ 1.1 at peak in every zone
Supply-short zone-hours %	Share of peak zone-hours with ratio < 1	—	≤ 10%
Unmet demand / peak	placed – delivered, peak blocks	—	minimise
Spatial-mismatch index	corr. of (idle ratio in zone A) with (unmet demand in adjacent zone B), same hour	—	monitor; the incentive should reduce it
ETA gap (short vs long zones)	ETA p90 short-tier – ETA p90 long-tier	—	narrowing

9.4 Courier economics & unit economics (page 4)

KPI	Definition	Benchmark	Target
Courier earnings per active hour (BRL)	Payout incl. tips & bonus ÷ active hours	R\$ 22–34	≥ R\$ 26 at peak
Utilisation %	Active ÷ logged-in minutes	55%–80%	≥ 70% at peak
Payout per delivery (BRL)	Total payout ÷ deliveries	—	context
Contribution margin per delivered order (BRL)	Take (commission + fee share) – payout – payment & support cost	R\$ 6–14	≥ R\$ 9
Incentive spend (BRL)	Total dynamic-bonus payout	—	≤ budget
Incentive cost per incremental delivered order (BRL)	Bonus spend ÷ incremental delivered orders	—	< contribution margin (≈ R\$ 9–11)
Incentive spend as % of courier payout	Bonus ÷ total payout	—	≤ 8%
Supply elasticity to incentive	% change in available courier-hours in a zone-block per R\$1 of bonus	—	quantify (Phase 10)

9.5 Experiment scorecard (Pricing & Incentive Experiment page)

KPI	Definition
Fulfillment rate by arm (+ lift, cluster-robust CI, p, RI p)	Primary metric, \$5.3
Weekly fulfillment lift across the 10 weeks	Novelty / decay diagnostic, \$5.5
Balance check (SRM χ^2 + SMD / Love plot)	Randomization check
Incentive cost per incremental delivered order (+ CI)	Guardrail G1 — the go/no-go economic test
ETA p50 / p90 by arm	Guardrail G2
Cancellation rate by arm and by cause	Guardrail G3
Courier earnings per active hour by arm	Guardrail G4
Neighbour-zone spillover: Δ fulfillment & Δ idle ratio in adjacent control zones	Guardrail G5 — cannibalisation
Effect by supply-stress tier and by zone (+ arm × tier interaction test)	Heterogeneity, \$5.6
Metro-level net lift vs summed local lift	Cannibalisation-adjusted bottom line

10. Targets and plan

- The plan is built on the **same basis as the actuals** (same fulfillment, ETA, cancellation and contribution-margin definitions).
- **Plan fulfillment rate:** ≥ 98% metro-wide, ≥ 95% in every peak zone-hour.
- **Plan p90 ETA:** ≤ 55 min at peak in every zone.
- **Plan supply-short peak zone-hours:** ≤ 10%.
- **Plan incentive efficiency:** cost per incremental delivered order **below** the contribution margin of a delivered order.
- "vs Target" measures exist per KPI; **ETA, cancellation rate, idle ratio, no-courier rate, courier CAC and incentive cost per incremental order are scored lower-is-better.**

11. Multi-currency handling

Out of scope by design. All monetary values are **BRL**. No *_local / *_usd split, no FX table, no reporting-currency slicer. Multi-currency was the VoltEdge project's job; repeating it here would only distract from the marketplace analysis.

12. Out of scope (explicit)

- A second city, expansion modelling, multi-currency / FX.
- Restaurant-side economics (commission optimisation, menu pricing, prep-time SLAs) beyond a fixed input to demand and unit economics.
- Consumer-side pricing experiments (delivery fee, coupons, subscription) — this project's intervention is **supply-side only**.
- Dispatch / routing-algorithm design and real-time optimisation.
- A courier-churn or demand-forecast **model**. Phase 10 quantifies supply elasticity and the experiment effect; it does not ship a predictive scorecard.
- Fraud (GPS spoofing, incentive gaming, collusion) beyond a noted data-quality tail.
- Weather/traffic modelling — rain and events enter only as covariate flags.
- Real personal data. Every courier, customer, restaurant, order, GPS ping and payout is synthetic and seed-generated.
- The **true incentive effect size and the cannibalisation term** — decided in Phase 3, detected blind in Phase 10.

13. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	20 months, 2025-01-01 → 2026-08-31; data-gen date 2026-09-10; mature operation throughout (no launch ramp).
A2	Geography	One metro (Belo Horizonte), 12 zones, fixed adjacency graph, one mid-window boundary redraw as a DQ issue.
A3	Baseline supply-stress tier	short / balanced / long, fixed from months 1–15 liquidity; ~4 / 5 / 3 zones.
A4	Two marketplace sides	Supply = available courier-hours; demand = demanded delivery-hours; never conflated.
A5	Active courier	≥ 1 completed delivery in trailing 7 days (weekly) / 28 days (monthly).
A6	Liquidity ratio	available courier-hours ÷ demanded delivery-hours, per zone × hour.
A7	Primary metric	Fulfillment rate = delivered ÷ placed, zone-day peak block.
A8	ETA	Placed → delivered minutes; p50 and p90 both reported.
A9	Cancellation taxonomy	customer · courier · no-courier · restaurant.
A10	Experiment unit / split	Zone-day , 50/50, stratified by zone × supply-stress tier, balanced on day-of-week; peak block 18:00–22:00.
A11	Experiment window	10 weeks, 2026-04-06 → 2026-06-14 (window-months 16–18); post-window to 2026-08-31.
A12	Treatment	Flat per-delivery bonus (R\$, in <code>etl/config.py</code>) for peak-block deliveries in the treated zone + in-app surge-zone push.
A13	Primary decision rule	Positive, significant ($\alpha = 0.05$, cluster-robust), ≥ MDE, and guardrail G1 (cost per incremental order) below the delivered-order contribution margin.
A14	Guardrails	G1 incentive cost / incremental order · G2 ETA p50/p90 · G3 cancellation rate by cause · G4 courier earnings / active hour · G5 neighbour-zone spillover.
A15	MDE / power	Computed for the zone-day unit (~420/arm), ICC ≈ 0.15, design effect ≈ 1.9; pooled MDE ≈ +1.3 pp, supply-short-tier MDE ≈ +2.2 pp; RI + wild-cluster bootstrap cross-checks. Finalized in Phase 10 from realized variance.
A16	Cannibalisation	A real, modest negative spillover to adjacent control zones baked into the data in Phase 3; Phase 10 estimates its sign and size and reports a cannibalisation-adjusted metro net lift.
A17	Required diagnostics	SRM/balance, covariate Love plot, novelty decay, tier & zone heterogeneity + interaction test, spillover estimate, persistence.
A18	Currency	BRL only; no FX in scope.
A19	Seasonality	Rainy-season and June-festas overlap the window edges; Carnaval deliberately excluded; concurrent control arm handles metro-wide shocks.
A20	True treatment effect	Deliberately not decided here; set in ETL (Phase 3), detected blind in Phase 10.

End of Phase 1 deliverable.

SwiftBite Delivery — Data Model

Document status: Phase 4 deliverable. **Depends on:** 02_data_architecture.md , 03_data_dictionary.md (auto-generated). Describes the curated star as it will be imported into the Power BI semantic model (Phase 6).

PT. O star curado: matriz de barramento (fatos × dimensões), grão e medidas de cada fato, atributos de cada dimensão, e as relações físicas para o modelo semântico.

1. Bus matrix (facts × dimensions)

Fact ↓ / Dim →	date	time_block	zone	courier	customer	restaurant	incentive_campaign
fact_order	•	•	• (dropoff current; +asbooked, pickup inactive)	•	•	•	(via zone+date)
fact_delivery	• (dropoff_ts)	•	•	•	(via order)	(via order)	(via zone+date)
fact_courier_shift_block	•	•	•	•			(via zone+date)
fact_zone_hour	•	•	•				(via zone+date)
fact_incentive_assignment	•		•				•
fact_experiment_zone_day	•		•				•
fact_experiment_zone_block_week	•	•	•				•

• = direct relationship; (via ...) = reachable through another table, not a physical relationship. Every fact carries date_key (int YYYYMMDD) → dim_date[date_key] (day grain, unique) and, where relevant, time_block_key (0–23) → dim_time_block.

2. Fact tables — grain & measures

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_order	one placed order	basket_value_brl , delivery_fee_brl , count	outcome , eta_min (delivered only), assignment_latency_s , trip_distance_km , is_peak_block , on_treated_zone_day , is_adjacent_to_treated_zone_day
fact_delivery	one delivered order	trip_min , trip_distance_km , courier_base_payout_brl , tip_brl , incentive_bonus_brl , total_payout_brl	on_treated_zone_day , experiment_phase
fact_courier_shift_block	courier × zone × date × hour	logged_in_min , active_min , idle_min , cooldown_min , deliveries , base_payout_brl , tip_brl , bonus_brl , earnings_brl	is_peak_block
fact_zone_hour	zone × date × hour	orders_placed , orders_delivered , cancelled_{customer,courier,no_courier,restaurant} , unmet_demand , demanded_delivery_hours , available_courier_hours , idle_courier_hours	fulfillment_rate , eta_p50_min , eta_p90_min , assignment_latency_p90_s , liquidity_ratio , idle_courier_ratio , active_couriers , is_rainy , has_local_event , experiment_arm
fact_incentive_assignment	experiment zone-day (~840)	count	arm , bonus_brl_per_delivery , stratum , day_of_week , n_adjacent_zones_treated , is_adjacent_to_treated , pre_fulfillment_rate , pre_eta_p90_min , pre_orders_placed_mean , pre_liquidity_ratio , pre_idle_courier_ratio
fact_experiment_zone_day	experiment zone-day (~840)	orders_placed , orders_delivered , deliveries , cancelled_no_courier , available_courier_hours , incentive_spend_brl	fulfillment_rate (primary), eta_p50_min , eta_p90_min , cancel_rate , cancel_no_courier_rate , courier_earnings_per_active_hour_brl , incentive_cost_per_delivered_order_brl , active_couriers , arm , stratum ,

Fact	Grain	Additive measures	Non-additive / semi-additive
			n_adjacent_zones_treated , is_adjacent_to_treated
fact_experiment_zone_block_week	zone × experiment week × peak hour- block	orders_placed , orders_delivered , available_courier_hours	fulfillment_rate , eta_p90_min , arm

fact_experiment_zone_day and fact_incentive_assignment share the (zone_key , date_key) key 1:1 — assignment carries the design + frozen covariates, the outcome table carries the realised metrics. Phase 10 joins them; the report reads the outcome table directly.

3. Dimensions — key attributes

Dimension	Key	Attributes used in the report
dim_date	date_key (int YYYYMMDD)	year , quarter , month , month_name , iso_week , day_of_week , is_weekend , is_holiday_br , is_rainy_season , experiment_phase , experiment_week
dim_time_block	time_block_key (hour 0-23)	hour , daypart , is_peak_block , block_label
dim_zone	zone_key	zone_id , zone_name , macro_area , area_km2 , centroid_lat/lon , n_neighbours , baseline_supply_stress_tier , avg_trip_km_baseline , is_boundary_redrawn
dim_zone_adjacency	(zone_key , neighbour_zone_key)	bridge — feeds the cannibalisation / spillover measures
dim_courier	courier_key	signup_cohort_month , vehicle_type , acquisition_channel , home_zone_key , is_test_account
dim_customer	customer_key	customer_segment , home_zone_key
dim_restaurant	restaurant_key	zone_key , cuisine , price_band
dim_incentive_campaign	campaign_key (also unique on zone_key + date_key)	arm , bonus_brl_per_delivery , push_radius_m , peak_block_start_hour , peak_block_end_hour , stratum , day_of_week

Sentinel keys. courier_key = -1 (no courier ever assigned — the cancelled_no_courier path), -2 (orphan courier id not in the master). Facts are never dropped for a missing dimension row.

4. Relationships

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_order	placed_date_key	dim_date	date_key	*:1	single	yes
fact_order	time_block_key	dim_time_block	time_block_key	*:1	single	yes
fact_order	dropoff_zone_key_current	dim_zone	zone_key	*:1	single	yes
fact_order	dropoff_zone_key_asbooked	dim_zone	zone_key	*:1	single	inactive
fact_order	pickup_zone_key	dim_zone	zone_key	*:1	single	inactive
fact_order	customer_key	dim_customer	customer_key	*:1	single	yes
fact_order	restaurant_key	dim_restaurant	restaurant_key	*:1	single	yes
fact_order	courier_key	dim_courier	courier_key	*:1	single	yes
fact_delivery	order_key	fact_order	order_key	1:1	single	yes (or merged)
fact_delivery	courier_key	dim_courier	courier_key	*:1	single	yes
fact_delivery	zone_key	dim_zone	zone_key	*:1	single	yes
fact_courier_shift_block	courier_key	dim_courier	courier_key	*:1	single	yes
fact_courier_shift_block	zone_key	dim_zone	zone_key	*:1	single	yes
fact_courier_shift_block	date_key / time_block_key	dim_date / dim_time_block	key	*:1	single	yes
fact_zone_hour	zone_key	dim_zone	zone_key	*:1	single	yes

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_zone_hour	date_key / time_block_key	dim_date / dim_time_block	key	*:1	single	yes
fact_incentive_assignment	campaign_key	dim_incentive_campaign	campaign_key	*:1	single	yes
fact_incentive_assignment	zone_key / date_key	dim_zone / dim_date	key	*:1	single	yes
fact_experiment_zone_day	zone_key / date_key	dim_zone / dim_date	key	*:1	single	yes
fact_experiment_zone_block_week	zone_key / date_key / time_block_key	dim_zone / dim_date / dim_time_block	key	*:1	single	yes

dim_zone_adjacency is a **bridge**: dim_zone[zone_key] 1 : * dim_zone_adjacency[zone_key] , and dim_zone_adjacency[neighbour_zone_key] is used by measures (not a physical relationship) to sum a metric over a zone's neighbours for the spillover guardrail.

fact_order carries three zone roles; only **dropoff (current)** is the active relationship. dropoff (as-booked) and pickup are inactive relationships activated with USERELATIONSHIP inside specific measures (as-booked drift; pickup-zone supply).

```
erDiagram
    dim_date ||--o{ fact_order : date
    dim_time_block ||--o{ fact_order : hour
    dim_zone ||--o{ fact_order : "dropoff (current)"
    dim_customer ||--o{ fact_order : places
    dim_restaurant ||--o{ fact_order : from
    dim_courier ||--o{ fact_order : delivers
    fact_order ||--|| fact_delivery : "delivered"
    dim_zone ||--o{ fact_zone_hour : ""
    dim_date ||--o{ fact_zone_hour : ""
    dim_time_block ||--o{ fact_zone_hour : ""
    dim_courier ||--o{ fact_courier_shift_block : ""
    dim_zone ||--o{ fact_courier_shift_block : ""
    dim_incentive_campaign ||--|| fact_incentive_assignment : "zone-day"
    fact_incentive_assignment ||--|| fact_experiment_zone_day : "zone-day"
    dim_zone ||--o{ fact_experiment_zone_day : ""
    dim_zone ||--o{ dim_zone_adjacency : "bridge"
```

5. Measure groups (Phase 7 preview)

Marketplace Health (fulfillment, ETA p50/p90, cancellation by cause, assignment latency) · Liquidity (available vs demanded courier-hours, liquidity ratio, idle-courier ratio, unmet demand, spatial-mismatch index) · Supply & Couriers (active couriers, CAC, retention, deliveries/active-hour) · Courier Economics (earnings/active-hour, utilisation, contribution margin, incentive spend, **incentive cost per incremental delivered order**) · **Experiment** (fulfillment by arm + lift + cluster-robust CI + p + RI p, weekly series, balance, guardrails G1–G5, effect by tier + interaction, neighbour-zone spillover, cannibalisation-adjusted net) · Time Intelligence (WoW, trailing-4-week, YoY context) · Targets vs Plan · a dynamic Exec Insight narrative measure.

End of Phase 4 deliverable (Data Model).

SwiftBite Delivery — Data Quality

Document status: Phase 4 deliverable. **Depends on:** `02_data_architecture.md`, `04_etl_pipeline.md`. **Companion artifact:** `data/quality/dq_report.md` (regenerated by `python etl/dq_checks.py`; the one data file committed to the repo).

PT. Como a sujeira é injetada de propósito, onde é tratada, e o que o build recusa a deixar passar. Oito classes (Q01–Q08), cada uma com um contador no `dq_report.md` e, quando é um invariante, uma asserção dura em `dq_checks.py`.

1. Philosophy

Every messy trait is **injected on purpose** by `01_generate_raw.py`, **handled** by `02` (parse / repair) or `03` (conform / sentinel), **counted** into a per-entity DQ fragment, and — where it is an invariant — **asserted** in `dq_checks.py`, which fails the build. Raw is never edited after generation. One `SEED` makes the whole thing reproducible, so the counters below are stable.

2. The eight injected classes

#	Issue	Injected in	Handled in	Hard check
Q01	Mixed datetime formats — <code>DD/MM/YYYY HH:MM</code> vs ISO in <code>OrderCore</code> ; Unix-epoch ms in the GPS reference; Excel serials for PayHub settlement weeks	<code>ordercore</code> , <code>trace</code> , <code>payhub</code>	<code>02</code> split parser (<code>^\d{4}-</code> → ISO fast path, else <code>BR</code>); <code>from_excel_serial</code>	<code>datetimes_unparseable == 0</code>
Q02	Timezone drift — PayHub & Boost export UTC <code>Z</code> with no offset; <code>OrderCore</code> & <code>RiderApp</code> are Brazil-local with no marker	<code>payhub</code> , <code>boost</code> vs <code>ordercore</code> , <code>riderapp</code>	<code>02</code> localises the naive feeds <code>America/Sao_Paulo</code> → UTC, parses the <code>Z</code> feeds as UTC; <code>03</code> derives the local calendar day/hour for keys	no naive timestamps in curated; peak-block bucketing exact
Q03	Money as locale strings — <code>"R\$ 62,90"</code> , <code>"1.234,56"</code> , <code>"(12,50)"</code> (reversal)	<code>ordercore</code> basket, <code>payhub</code> amounts	<code>02</code> <code>parse_money</code> (strip <code>R\$</code> /thousands, <code>,</code> → <code>.</code> , parens → negative)	no non-null money column is object dtype; payout / basket ≥ 0
Q04	GPS noise — lat/long jitter, far-away cold-start pings, <code>(0,0)</code> null-island, irregular cadence	<code>trace</code> <code>location_pings</code> (7-day reference)	<code>02</code> <code>bbox</code> + null-island drop, resample to courier-minute state	0 out-of-bbox pings survive into <code>stg_trace_ping_minute</code>
Q05	Duplicate rows — mobile retry dupes on orders; at-least-once status / offer / bonus logs; a re-delivered PayHub month	<code>ordercore</code> , <code>riderapp</code> , <code>dispatch</code> , <code>payhub</code>	<code>02</code> exact-dedup; drop the re-delivered file on <code>order_id</code>	<code>fact_order</code> / <code>fact_delivery</code> key-unique; <code>fact_experiment_zone_day</code> unique on (<code>zone_key</code> , <code>date_key</code>)
Q06	Orphan keys — a slice of <code>offers</code> reference a <code>courier_id</code> absent from the master	<code>riderapp</code>	<code>03</code> sentinel <code>courier_key = -2</code> , counted	every fact FK resolves to a real or sentinel key
Q07	Zone boundary redraw (window-month 9) — the as-booked label is stale for the two re-cut zones	<code>georef</code> + <code>ordercore</code> label	<code>03</code> keeps both <code>dropoff_zone_key_asbooked</code> and <code>_current</code> (centroid-resolved only for the redrawn pair pre-redraw); experiment/zone-hour facts key off <code>_current</code>	every order has a non-null <code>dropoff_zone_key_asbooked</code> ; drift reported
Q08	Encoding damage — Latin-1 mojibake in address <code>bairro</code> / <code>logradouro</code> and restaurant names	<code>ordercore</code>	<code>02</code> <code>ftfy.fix_text</code> on the text columns	no <code>Ã</code> / <code>Â</code> mojibake sequences remain in curated address / restaurant text

Also handled without ceremony: `courier_id` casing → upper-normalised; post-midnight Boost campaign edits → earliest (`00:00` -fixed) row per zone-day; the `no_courier_found` → late-assignment funnel → one reconciled outcome per order; `status` / offer `outcome` / cuisine free-text → controlled vocab in `etl/vocab/`.

3. Operational identities asserted (hard)

- Order conservation**, every zone-hour: `orders_placed == orders_delivered + Σ cancelled_{customer, courier, no_courier, restaurant}` (exact).
- Rate sanity**: `fulfillment_rate ∈ [0, 1]`; `eta_p50_min ≤ eta_p90_min`; `liquidity_ratio ≥ 0`; `idle_courier_ratio ∈ [0, 1]`.

- **Experiment integrity:** every `fact_experiment_zone_day` zone-day exists in `fact_incentive_assignment` ; treated zone-days have `bonus_br1_per_delivery > 0` , control `== 0` ; ≥ 75 days of post-experiment fact data.

4. Benchmark bands checked (soft — reported, not fatal)

On the **pre-experiment baseline** (window-months 1–15) the following must sit mid-band, or the calibration is off (see `01_business_context_kpis.md` §9):

Metric	Band	Current build
Fulfillment rate	0.93 – 0.995	0.956 ✓
ETA p50 (min)	24 – 42	≈ 38 ✓
ETA p90 (min)	42 – 80	≈ 46 ✓
Cancellation rate	0.01 – 0.07	0.044 ✓
Idle-courier ratio	0.10 – 0.40	0.286 ✓
Courier earnings / active-hour (BRL)	20 – 40	≈ 33 ✓
Per-stratum arm allocation	0.45 – 0.55	0.50 all 12 ✓

The experiment arms are allowed to diverge inside the window — that is the point.

5. Current status

`python etl/dq_checks.py` → **13 / 13 hard checks pass · 0 soft warnings**. The full counter table (rows in / rows out, dupes dropped, money strings parsed, GPS pings dropped, mojibake repaired, late edits reconciled, ...) is in [data/quality/dq_report.md](#) .

End of Phase 4 deliverable (Data Quality).

SwiftBite Delivery — Report Structure

Document status: Phase 8 deliverable. The report is built from code: `powerbi/gen_report_visuals.py` writes `powerbi/_report_layout.json`, `powerbi/_drive_pbir.py` drives the `pbir` CLI to produce `powerbi/SwiftBiteDelivery.Report/**`. `powerbi/build_report.sh` orchestrates the whole chain. `pbir validate --all` passes.

PT. 4 páginas, uma pergunta de negócio por página. Cada página: banda verde com o wordmark, hairline tangerina, título + a pergunta, uma linha de 6 KPI cards numa moldura branca, e 4 visuais de conteúdo. 3 slicers globais (período / zona / tier de estresse de oferta).

Every page

- **Header band** (green-deep, 58 px) with the *SwiftBite* wordmark, a tangerine hairline, and a right-aligned "Marketplace Analytics" label — carried by the page-background PNG (`assets/gen_logo.py`).
- **Title + business question** on the light body.
- **Slicer row:** Analysis period (`dim_date[date]`) · Zone (`dim_zone[zone_name]`) · Supply-stress tier (`dim_zone[baseline_supply_stress_tier]`).
- **KPI frame:** a white rounded shape holding **6 value cards**.
- **Two content rows** on the 3-30-300 gradient (4 visuals).

All numbers come from the 69 explicit DAX measures in `_Measures` (`discourageImplicitMeasures` is on); see `06_dax_measures.md`.

1 · Executive Overview

Question — Is the marketplace healthy, and did the zone-hour incentive earn its cost?

KPIs	Visuals
Fulfillment Rate · Peak ETA p90 · No-Courier Rate · Idle Courier Ratio · Incentive Lift (pp) · Cost / Incremental Order	fulfillment-rate trend (line) · Exec Insight dynamic narrative card · fulfillment lift by supply-stress tier (bar) · weekly fulfillment treated vs control (line)

2 · Marketplace Health

Question — Where and when does liquidity break?

KPIs	Visuals
Orders Placed · Fulfillment Rate · ETA p90 · No-Courier Rate · Liquidity Ratio · Idle Courier Ratio	fulfillment rate by zone (bar) · liquidity ratio by zone × daypart (matrix) · ETA p90 by hour of day (line) · cancellations by cause and zone (column)

3 · Pricing & Incentive Experiment

Question — Is the incentive real, safe on the guardrails, and not just cannibalising neighbour zones?

KPIs	Visuals
Fulfillment Lift (pp) · ETA p90 Lift (min) · No-Courier Lift (pp) · Cost / Incremental Order · Neighbour-Zone Gap (pp) · Adjusted Net Lift (pp)	fulfillment lift by tier — heterogeneity (bar) · fulfillment by arm (column) · weekly fulfillment treated vs control — launch bump vs settled (line) · guardrails by arm (table)

4 · Courier Economics

Question — What do couriers earn, how utilised is supply, and how elastic is it to the bonus?

KPIs	Visuals
Active Couriers · Earnings / Active Hr · Utilisation · Deliveries / Active Hr · Contribution Margin · Incentive % of Payout	earnings per active hour by daypart (column) · utilisation by zone (bar) · available vs demanded courier-hours by hour (line) · incentive cost per delivered order by tier (column)

Layout warnings from `pbir validate` (`VISUAL_OVERLAP` / `VISUAL_UNDSIZED`) are expected — the KPI value cards sit inside the frame shape by design; final pixel spacing is a Power BI Desktop polish pass.

